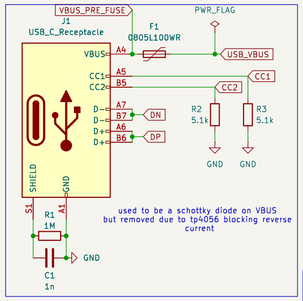
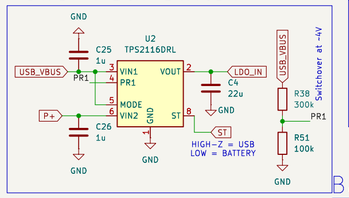
USB-C Connector

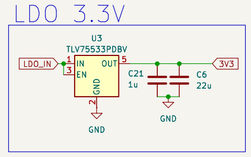


USB-C ESD Protection

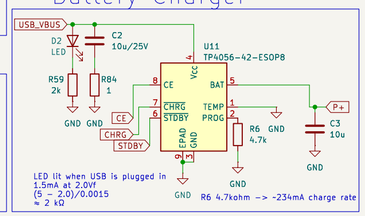
Battery<->USB Power Path Switch



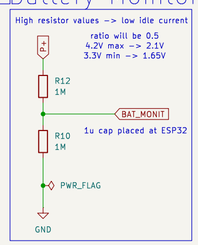
LDO 3.3V



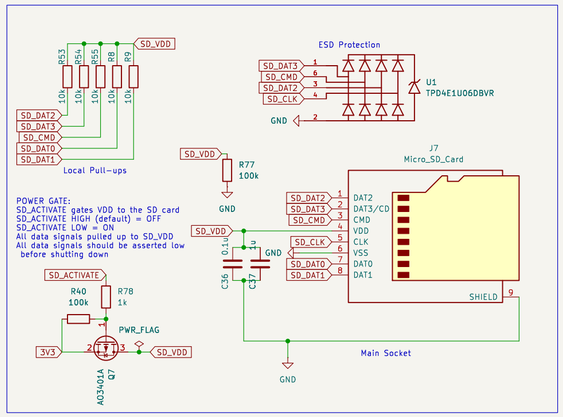
Battery Charger



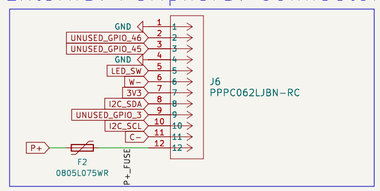
Battery Monitor



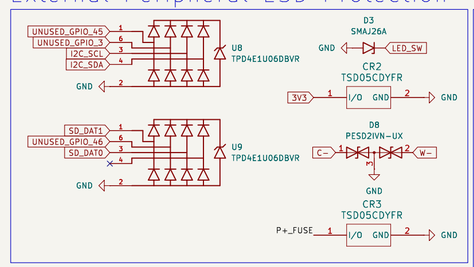
SDMMC Interface



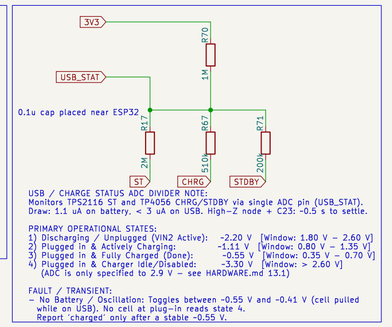
External Peripheral Connector



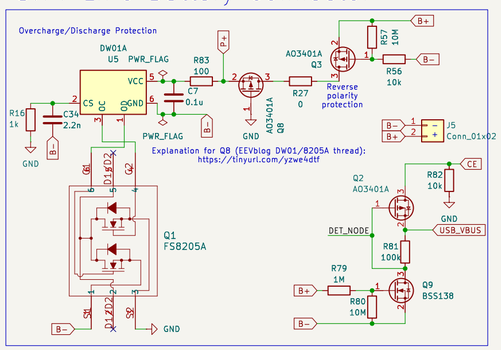
External Peripheral ESD Protection



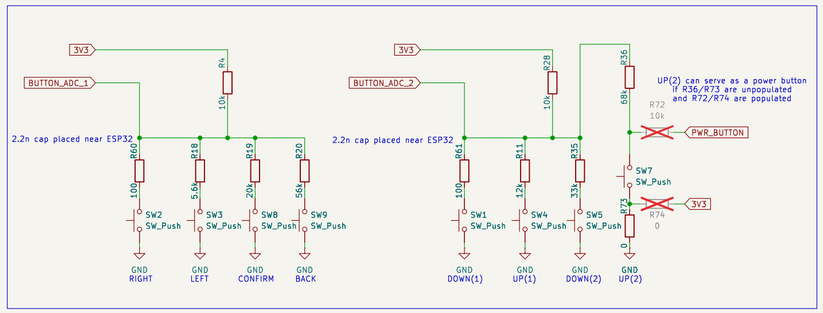
USB Status



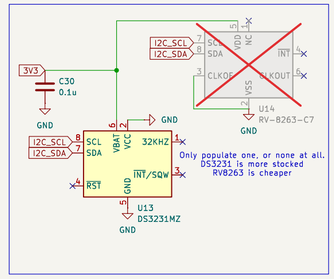
3.7V LiPo Battery Connector



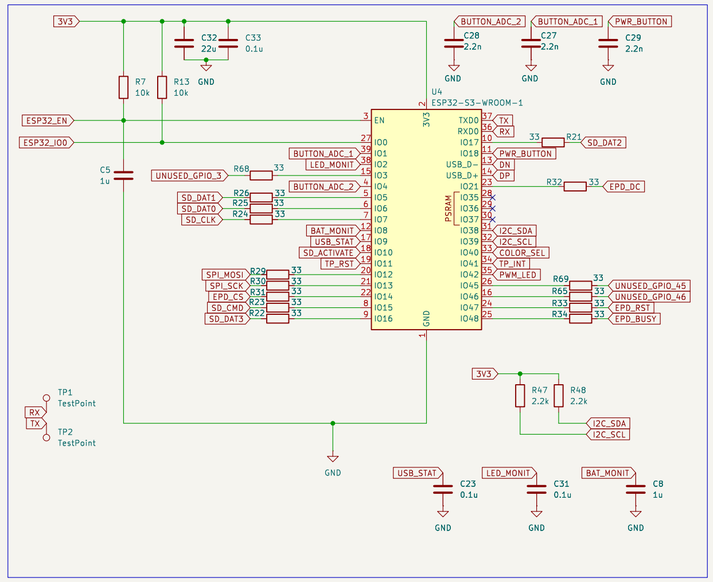
Interface Buttons



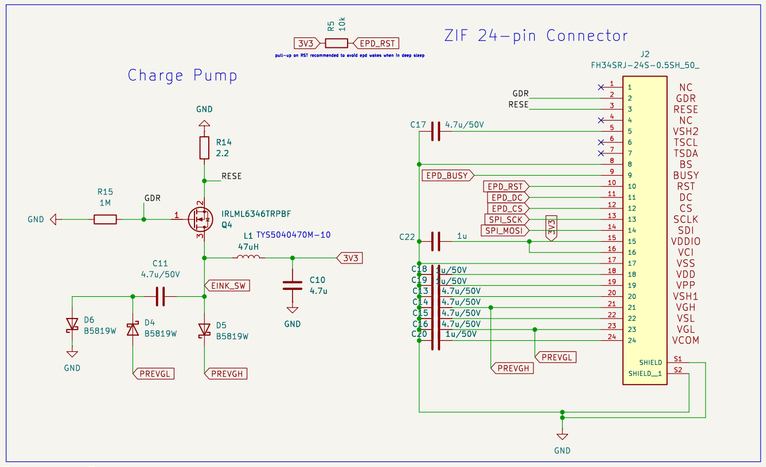
External RTC



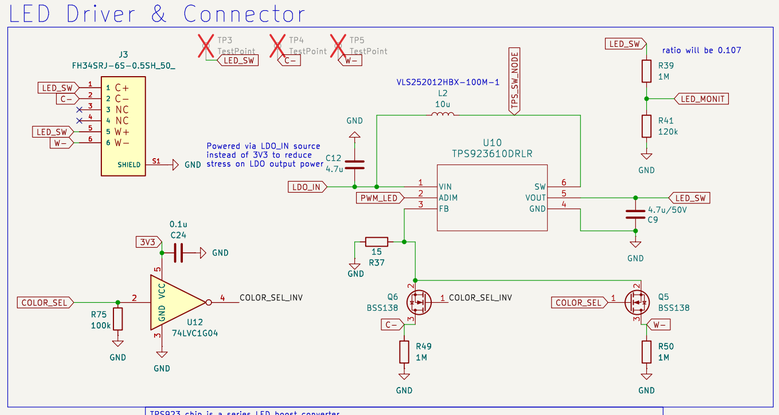
ESP32 Pin-out



E-ink Driver & Connector

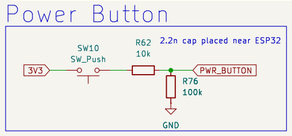


LED Driver & Connector

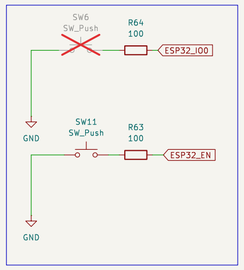


TPS923 chip is a series LED boost converter.
Recommended to run between 10kHz–25kHz via PWM_LED for brightness control.
PWM_LED (ADIM) is also the ENABLE: first HIGH pulse must be >= 40 us (tADIM_EN), so hold HIGH >= 100 us before starting the PWM carrier; LOW for > 2.5 ms (tADIM_SD) shuts the driver down.
Return path for LEDs is toggled via COLOR_SEL for warm/cool selection.
Feedback current is ~13.3mA tuned via R37. Reference datasheet to change as needed.
LED_MONIT is analog signal that references 0–24.5V output on LED voltage. (ADC range from 0 to ~30V)
Internal OVP should cut out at ~24V, but software protection should kick in and disable PWM_LED at safe voltage above known forward voltage (15V)
After 3 OVP trips the driver LATCHES off: re-applying PWM will not restart it – hold PWM_LED LOW > 2.5 ms (or power-cycle).
Change COLOR_SEL only with PWM duty at 0. With no light connected, LED_MONIT (software trip ~23V) and the U10 OVP latch protect the driver; avoid enabling it with J3 and J6 empty.

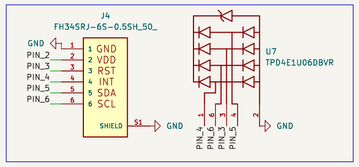
Power Button



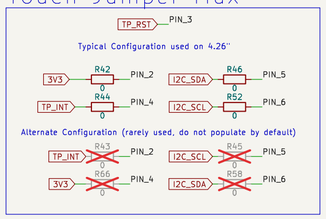
Bootloader Buttons



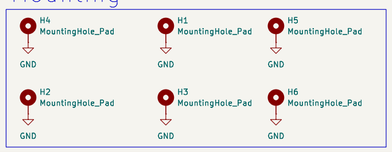
Touch Connector + ESD



Touch Jumper Mux



Mounting



github.com/ianchasse/silkscreen-pcb
© 2026 idc LLC
Licensed under CERN-OHL-S-2.0
Open-hardware e-reader base board

idc LLC

Sheet: /
File: silkscreen_pcb.kicad_sch

Title: Silkscreen Reader PCB

Size: A2	Date: 2026-10-22	Rev: 1.02
KiCad E.D.A. 9.0.6		Id: 1/1